# Mobile Robotics Lab Manual

# Lab Session 8: Building and Visualizing a Custom Mobile Robot using URDF

## Objective

In this lab, students will begin constructing a mobile robot using the Unified Robot Description Format (URDF). The focus will be on designing the base structure of a differential drive robot and visualizing it in RViz using ROS 2 tools. The simulation of this robot in Gazebo will be covered in the next session.

## Learning Outcomes

By the end of this lab session, students will be able to:

- Understand the structure and purpose of URDF files.
- Create and organize a ROS 2 package for a robot description.
- Build a simple mobile robot model with links and joints.
- Visualize their robot in RViz.
- Modify and extend the robot with their own designs.

## Key Tools and Packages

- **URDF (Unified Robot Description Format):** An XML format used to describe the physical configuration of robots, including links, joints, and sensors.
- **urdf_tutorial:** A ROS 2 package that provides example URDF files and a basic launch setup to visualize URDFs in RViz.
- **tf2_tools:** A utility package in ROS 2 that provides tools to visualize and inspect transformations between coordinate frames.
- **robot_state_publisher:** A ROS 2 node that reads the URDF and publishes coordinate transforms (`tf`) for all the robot links based on joint states.

## Pre-Lab Setup

### Install Required Packages

```
sudo apt install ros-humble-urdf-tutorial
sudo apt install ros-humble-tf2-tools
```

## Step-by-Step Lab Guide

### Task 1: Create URDF Folder Structure

```
cd ~/ros2_ws/src/my_robot_description
mkdir urdf launch rviz
```

### Task 2: Write a Basic URDF File

Create a file named `my_robot.urdf` inside the `urdf/` folder, a sample code is shown below:

```
<robot name="my_mobile_robot">
  <link name="base_link">
    <visual>
      <geometry>
        <box size="0.4 0.3 0.1"/>
      </geometry>
      <material name="blue"/>
    </visual>
  </link>


  <link name="camera">
    <visual>
      <geometry>
        <cylinder length="0.05" radius="0.05"/>
      </geometry>
      <material name="black"/>
    </visual>
  </link>

  <joint name="base_camera_joint" type="fixed">
    <parent link="base_link"/>
    <child link="left_wheel"/>
    <origin xyz="0 0 0" rpy="0 0 0"/>
  </joint>

  </robot>
```

**Task 3: Build and Launch**

```
cd ~/ros2_ws
colcon build
source install/setup.bash
ros2 launch urdf_tutorial display.launch.py model:=<abs_path_to_urdf_file>
```

**Explore:** Move the sliders to simulate joint movement and observe the robot's change in RViz.

**TF Tree**

In a new terminal, run the node view_frames from the package tf2_tools to visualise frames attached to the robot links and joints.

```
ros2 run tf2_tools view_frames
```

## Student Task

After completing the guided tutorial with the provided URDF, each student must:

1. **Design a Custom Robot**
    - Add additional links (e.g., arms, sensors, decorative structures).
    - Use different geometries like spheres and cylinders.
    - Add more joints, experiment with joint types (fixed, continuous, revolute).
2. **Visualize in RViz**

    - Ensure your robot appears correctly.
    - Configure TF and RobotModel displays.
    - Set `base_link` as the fixed frame.
3. **Prepare for Simulation**
    - Think about how your robot will move in Gazebo.
    - Plan to add sensors (e.g., LiDAR) in the next session.

## Deliverables

- Screenshot of your custom robot visualized in RViz.
- Your final URDF file.
- Summary of what customizations you made.
- Optional: RViz configuration file (`.rviz`).